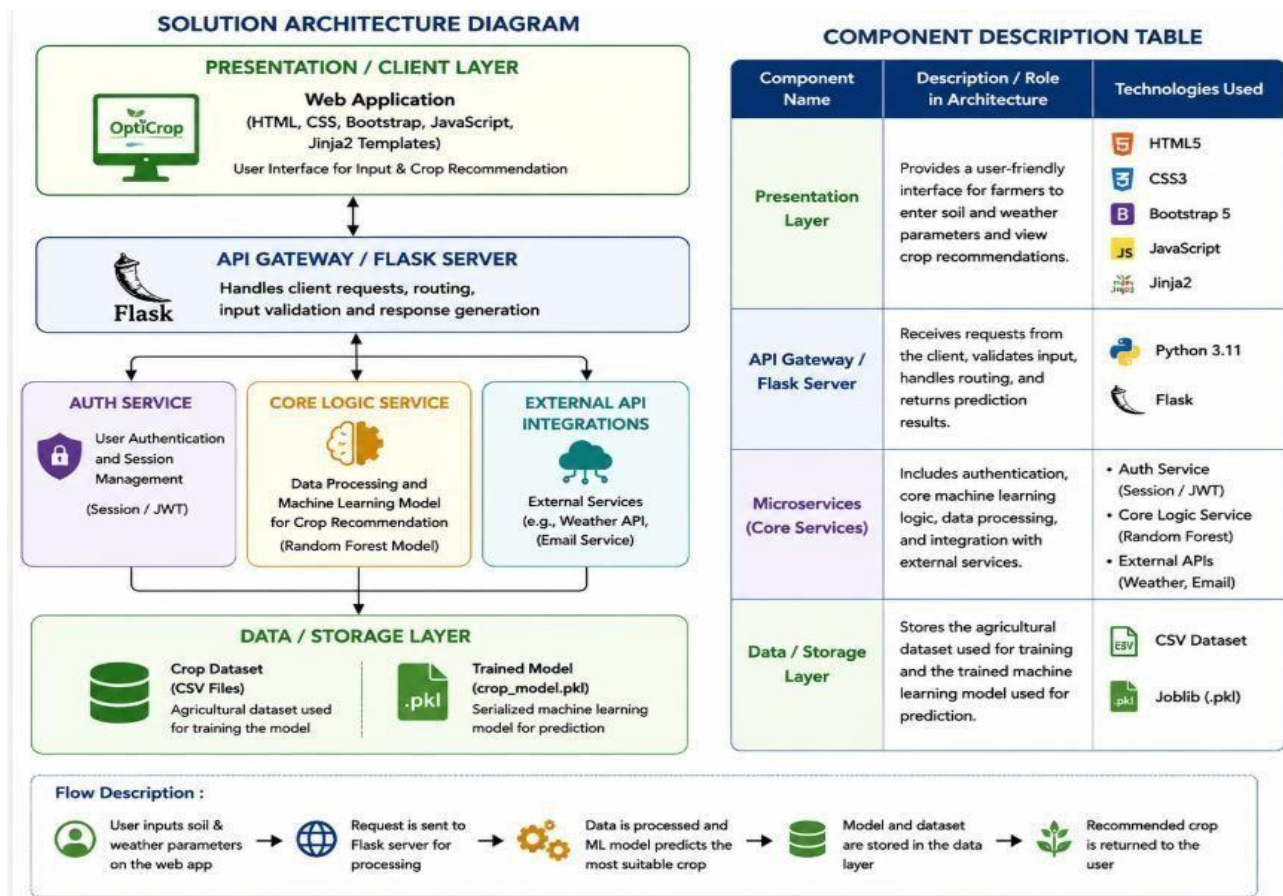


Solution Architecture

Date	01-Jul-2026
Team ID	XXXXXX
Project Name	OptiCrop: Smart Agricultural Production Optimization Engine
Maximum Marks	5 Marks

Solution Architecture Diagram:



Component Description Table

Component Name	Description / Role in Architecture	Technologies Used
Presentation Layer	Provides the user interface for farmers to enter soil and weather parameters and view crop recommendations.	HTML5, CSS3, Bootstrap 5, JavaScript, Jinja2
API Gateway / Application Layer	Receives user requests, validates inputs, communicates with the machine learning model, and returns prediction results.	Python 3.11, Flask
Core Logic Service (Machine Learning)	Processes input data, performs preprocessing, executes the trained Random Forest model, and generates the most suitable crop recommendation.	Scikit-learn, Pandas, NumPy, Joblib
Data / Storage Layer	Stores the crop recommendation dataset and the trained machine learning model used for prediction.	CSV Dataset, Joblib (.pkl) Model Files

- **Presentation Layer:** The Presentation Layer provides the user interface of the **OptiCrop** system, allowing farmers to enter soil nutrient values (Nitrogen, Phosphorus, Potassium) and environmental parameters such as temperature, humidity, pH, and rainfall. It displays the recommended crop and prediction results through a simple, responsive, and user-friendly web application.
- **Application Layer:** The Application Layer acts as the core processing unit of the system. It receives user inputs, validates the data, preprocesses the parameters, loads the trained machine learning model, performs crop prediction, and returns the recommended crop to the user. It also manages communication between the presentation layer and the data layer.
- **Data Layer:** The Data Layer stores and manages the Crop Recommendation Dataset, trained machine learning model (`crop_model.pkl`), preprocessing files, and other resources required for prediction. It supplies the necessary data to the application layer for accurate crop recommendations.
- **Overall Architecture:** The layered architecture of **OptiCrop** ensures smooth data flow, modular design, scalability, maintainability, and efficient integration of machine learning. It provides accurate crop recommendations while supporting future enhancements such as fertilizer recommendation, crop disease detection, yield prediction, weather API integration, and mobile application support.